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Inventor(s)	IGURA; Koushi et al.

CIRCUIT ASSEMBLY

Abstract

A circuit assembly includes a heat generating component that generates heat when energized; a energization member that connects to a connection portion of the heat generating component; a fastening member that fastens the energization member to the connection portion; and a heat capacity increasing component that thermally connects to a fastening site of the energization member and the connection portion and increases a heat capacity of the connection portion of the heat generating component.

Inventors: **IGURA; Koushi (Mie, JP), SHIMODA; Hiroki (Mie, JP), YANAGIDA; Taiji (Mie, JP)**

Applicant: **AUTONETWORKS TECHNOLOGIES, LTD.** (Mie, JP); **SUMITOMO WIRING SYSTEMS, LTD.** (Mie, JP); **SUMITOMO ELECTRIC INDUSTRIES, LTD.** (Osaka, JP)

Family ID: **78708548**

Assignee: **AUTONETWORKS TECHNOLOGIES, LTD. (Mie, JP); SUMITOMO WIRING SYSTEMS, LTD. (Mie, JP); SUMITOMO ELECTRIC INDUSTRIES, LTD. (Osaka, JP)**

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This is a Continuation of U.S. application Ser. No. 17/923,060, filed Nov. 3, 2022, which is a National Phase application of International Patent Application No. PCT/JP2021/018837, filed May 18, 2021, which claims priority to Japanese Patent Application No. 2020-089136, filed on May 21, 2020. The entire disclosure of each of the above-identified documents is incorporated herein by reference in its entirety.

TECHNICAL FIELD
[0002] The present disclosure relates to a circuit assembly including a heat generating component.
BACKGROUND ART
[0003] In a known circuit assembly provided with a heat generating component, such as a relay or fuse that generates heat when energized, a heat dissipating structure for dissipating the heat of the heat generating component may be provided. For example, in the structure described in Patent Document 1, heat generated by a relay housed in a case is dissipated using an intermediate portion of a bus bar that connects a connection portion of the relay and a connection terminal of a battery disposed outside the case. Specifically, a structure is described in which the intermediate portion of the bus bar, which extends outside the case housing the relay, is brought into contact with a chassis or a housing that houses an entire power supply apparatus via an insulative heat dissipation sheet, whereby heat generated by the relay is conducted to the chassis or the housing and dissipated.
CITATION LIST
Patent Documents
[0004] Patent Document 1: JP 2014-79093A
SUMMARY OF INVENTION
Technical Problem
[0005] In the structure of Patent Document 1, the heat dissipating structure is provided in the intermediate portion of the bus bar that forms an energization portion connecting the relay and the battery. Thus, though heat dissipation of the relay via the bus bar is promoted, there is a problem in that, when a large current flows, heat generated at the connection portion of the relay cannot be quickly reduced.
[0006] In view of this, a circuit assembly with a novel structure that can quickly reduce heat generation at a connection portion of a heat generating component is provided.
Solution to Problem
[0007] A circuit assembly of the present disclosure includes a heat generating component that generates heat when energized; a energization member that connects to a connection portion of the heat generating component; a fastening member that fastens the energization member to the connection portion; and a heat capacity increasing component that thermally connects to a fastening site of the energization member and the connection portion and increases a heat capacity of the connection portion of the heat generating component.
Advantageous Effects of Invention

[0008] According to the present disclosure, heat generation at a connection portion of a heat generating component can be quickly reduced.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. **1** is a perspective view illustrating a circuit assembly according to a first embodiment.

[0010] FIG. **2** is an exploded perspective view illustrating the circuit assembly illustrated in FIG. **1** in a state with the lid member forming the case removed.

[0011] FIG. **3** is an exploded perspective view of the circuit assembly illustrated in FIG. **1**.

[0012] FIG. **4** is a perspective view illustrating an energization member forming the circuit assembly illustrated in FIG. **1**.

[0013] FIG. **5** is a cross-sectional view taken along V-V in FIG. **2**.

[0014] FIG. **6** is a perspective view illustrating a circuit assembly according to a second embodiment and is an enlarged view of a main portion in a state with a lid member forming a case removed.

[0015] FIG. **7** is a cross-sectional view taken along VII-VII of FIG. **6**.

[0016] FIG. **8** is a perspective view illustrating a circuit assembly according to a third embodiment and is an enlarged view of a main portion in a state with a lid member forming a case removed.

[0017] FIG. **9** is a cross-sectional view taken along IX-IX in FIG. **8**.

[0018] FIG. **10** is a vertical cross-sectional view illustrating a circuit assembly according to a fourth embodiment and corresponds to FIG. **9**.

DESCRIPTION OF EMBODIMENTS

Description of Embodiments of Present Disclosure

[0019] Firstly, embodiments of the present disclosure will be listed and described. A circuit assembly of the present disclosure is [0020] (1) a circuit assembly including a heat generating component that generates heat when energized; a energization member that connects to a connection portion of the heat generating component; a fastening member that fastens the energization member to the connection portion; and a heat capacity increasing component that thermally connects to a fastening site of the energization member and the connection portion and increases a heat capacity of the connection portion of the heat generating component.

[0021] According to the circuit assembly of the present disclosure, the heat capacity increasing component that increases the heat capacity of the connection portion is provided at and thermally in contact with the fastening site of the connection portion corresponding to the heat generating site of the heat generating component and the energization member connected to the connection portion. Thus, heat at the connection portion transferred to the fastening site of the connection portion of the heat generating component and the energization member can be reduced by suppressing an increase in the temperature via the heat capacity increasing component thermally in contact with the fastening site. As a result, compared to a known structure in which an energization member is brought into contact with another member at a site separated from the connection portion of the heat generating component and heat is dissipated, heat generated at the connection portion of the heat generating component can be quickly reduce by the heat capacity increasing component. Note that the heat generating component includes a component that generates heat when energized, such as a relay, fuse, or the like.

[0022] The heat capacity increasing component may be any kind of component as long as the heat capacity of the connection portion of the heat generating component can be increased by thermally coming into contact with the heat capacity increasing component to the fastening site of the connection portion and the energization member. For example, a component made of metal with high thermal conductivity, such as iron, copper, aluminum, an alloy thereof, or the like or a

synthetic resin may be used. Also, the shape of the heat capacity increasing component is not particularly limited and may be any shape as long as the shape allows for the heat capacity increasing component to be in thermal contact with the fastening site of the connection portion and the energization member.
[0023] Any known fastening member may be used as the fastening member as long as it can be used to fasten the energization member, and advantageous examples include a bolt, a rivet, and the like. [0024] (2) Preferably, a heat conducting member and a case are further provided, wherein the energization member is thermally in contact with the case via the heat conducting member. The heat transferred to the energization member can be dissipated from the case via the heat conducting member. Thus, the heat of the heat generating component can be reduced. In the present aspect, preferably, a sheet-like heat conducting member is used. [0025] (3) Preferably, the heat capacity increasing component is made of metal; and the heat capacity increasing component is fastened to the connection portion together with the energization member via the fastening member. Because the heat capacity increasing component is made of metal and fastened to the connection portion together with the energization member, the heat capacity of the connection portion can be easily and reliably increased. Note that the heat capacity increasing component may be formed integrally with the energization member or may be a separate component from the energization member.
[0026] (4) Preferably, the heat capacity increasing component is overlapped with a surface on an opposite side to a contact surface of the energization member with the connection portion. Because the heat capacity increasing component is overlapped with the surface on the opposite side to the contact surface of the energization member with the connection portion, when the connection portion is fastened with the fastening member, the heat capacity increasing component is avoided from being disposed between the energization member and the connection portion. Thus, the heat capacity of the connection portion can be increased without increasing the electrical conduction resistance. [0027] (5) Preferably, the heat capacity increasing component is formed of an end portion of the energization member and is folded back and overlapped with a surface on an opposite side to a contact surface of the energization member with the connection portion. Because the heat capacity increasing component is formed of the end portion of the energization member, an increase in the number of parts can be suppressed. Also, because the heat capacity increasing component is folded back and overlapped with the surface on the opposite side to the contact surface of the energization member with the connection portion, when the connection portion is fastened with the fastening member, the heat capacity increasing component is avoided from being disposed between the energization member and the connection portion. Thus, the heat capacity of the connection portion can be increased without increasing the electrical conduction resistance.
[0028] (6) Preferably, the energization member includes a holding portion that holds the end portion of the energization member forming the heat capacity increasing component and the surface on the opposite side of the contact surface of the energization member with the connection portion in an overlapped state. With the holding portion, the gap between the energization member and the heat capacity increasing component (folded-back end portion of the energization member) can be kept small, and the energization member and the heat capacity increasing component can be stably brought into contact with one another across a wide contact area. Accordingly, the heat capacity of the connection portion can be more reliably increased. [0029] (7) Preferably, a coefficient of linear thermal expansion of the heat capacity increasing component ranges from ⅓ to 3 times a coefficient of linear thermal expansion of the fastening member. Because the heat capacity increasing component has a coefficient of linear thermal expansion similar to that of the fastening member, looseness of the fastening member caused by heat generation is unlikely to occur. [0030] (8) Preferably, the heat capacity increasing component and the fastening member are a same material. Because the coefficient of linear thermal expansion of the heat capacity increasing component and the fastening member is equal, looseness of the fastening member caused by heat generation can be more reliably suppressed. [0031] (9) Preferably, the heat capacity increasing

component is formed of a cap for installing on the fastening member. Even with a cap for installing on the fastening member, the heat capacity of the connection portion can be increased, and heat generation at a connection portion of a heat generating component can be quickly reduced. [0032] (10) Preferably, the cap is made of metal. By using the cap made of a metal with high thermal conductivity, an increase in the temperature at the connection portion can be suppressed. [0033] (11) Preferably, a heat conducting member is provided between the cap and the fastening member. Using the heat conducting member, heat can be stably transferred from the fastening member to the cap. In the present aspect, preferably, a grease-like heat conducting member is used, for example. [0034] (12) Preferably, the cap is made of synthetic resin. For example, by using a synthetic resin that is softer than metal as the material for the cap, the cap can be installed on the fastening member substantially without gaps. Thus, heat can be stably transferred from the fastening member to the cap, and the heat capacity of the connection portion can be more reliably increased.

DETAILS OF EMBODIMENTS OF PRESENT DISCLOSURE

[0035] Specific examples of the circuit assembly according to the present disclosure will be described below with reference to the drawings. Note that the present disclosure is not limited to these examples and is defined by the scope of the claims, and all modifications that are equivalent to or within the scope of the claims are included.

First Embodiment

[0036] The first embodiment of the present disclosure will be described below with reference to FIGS. **1** to **5**. A circuit assembly **10** of the first embodiment is installed in a vehicle (not illustrated), such as an electric vehicle or a hybrid vehicle, for example, and supplies and controls electric power from a power supply (not illustrated) such as a battery to a load (not illustrated) such as a motor. While the circuit assembly **10** can be oriented in any direction, in the following description, the X direction is defined as the forward direction, the Y direction is defined as the right direction, and the Z direction is defined as the upward direction. Also, for members including a plurality of members, a reference sign or number may only be given to one or more of the members of the plurality and not given to other members.

Circuit Assembly **10**

[0037] The circuit assembly **10** includes a relay **12**, which is an example of a heat generating component that generates heat when energized; an energization bus bar **16**, which is an example of an energization portion connecting to a connection portion **14** of the relay **12**; and a bolt **18**, which is an example of a fastening member that fastens the energization bus bar **16** to the connection portion **14** of the relay **12**. Also, the circuit assembly **10** includes a heat capacity increasing component **20** that is thermally in contact with a fastening site A (a region surrounded by a two-dot dash line in the diagrams) of the energization bus bar **16** and the connection portion **14**. The circuit assembly **10** further includes a case **22**. The relay **12**, the energization bus bar **16**, the bolt **18**, and the heat capacity increasing component **20** are all housed in the case **22**. The case **22** also houses a fuse **24** that generates heat when energized and a current sensor **26**.

Case **22**

[0038] The case **22** has an overall box-like shape and is made of a synthetic resin, for example. In the first embodiment, the case **22** is substantially rectangular, extending in the left-and-right direction in a plan view. The case **22** is divisible in the up-and-down direction and includes a base member **28** located on the lower side and a lid member **30** located on the upper side. The base member **28** has a box-like shape opening upward. The lid member **30** has a box-like shape opening downward. Also, the case **22** is formed by the upper opening portion of the base member **28** being covered by the lid member **30** and the base member **28** and the lid member **30** being fixed together. The method of fixing together the base member **28** and the lid member **30** is not limited, and various known fixing methods, such as adhesion, welding, press-fitting, protrusion-recess fitting, and the like can be used. Note that the case **22** may be made of metal, and the surface of the case **22** may be provided with an insulating cover to ensure it has insulating properties.

[0039] The base member **28** includes a bottom wall **32** with a substantially rectangular shape extending in the left-and-right direction and a peripheral wall **34** that projects upward from the outer peripheral edge portion of the bottom wall **32**. In the first embodiment, a first housing recess portion **36** with a rectangular shape opening upward is formed in the upper surface of the bottom wall **32**. In other words, a level difference **38** is formed on the upper surface of the bottom wall **32**, and the portion surrounded by the level difference **38** corresponds to the first housing recess portion **36**. In particular, in the first embodiment, at the upper surface of the bottom wall **32**, four first housing recess portions **36** are formed with a predetermined size and at a predetermined separation distance in the left-and-right direction.
[0040] As illustrated in FIG. **5**, a second housing recess portion **40** with a rectangular shape opening downward is formed at a position corresponding to the first housing recess portion **36** in the lower surface of the bottom wall **32**. In other words, a level difference **42** is formed on the lower surface of the bottom wall **32**, and the portion surrounded by the level difference **42** corresponds to the second housing recess portion **40**. Four second housing recess portions **40** are provided with a size and positions corresponding to the first housing recess portions **36**. Thus, in the first embodiment, at the positions where the first and second housing recess portions **36** and **40** are formed, the bottom wall **32** is thinner compared to other portions.
[0041] The lid member **30** includes an upper bottom wall **44** with a substantially rectangular shape extending in the left-and-right direction and a peripheral wall **46** that projects downward from the outer peripheral edge portion of the upper bottom wall **44**. In the first embodiment, opening portions **48***a* and **48***b* with a rectangular shape are formed in the left and right end portions of the upper bottom wall **44** extending through in the up-and-down direction. Also, a plurality of bolt insertion holes **50** are formed in the outer peripheral portion of the lid member **30** extending through in the up-and-down direction.
Relay **12**, Fuse **24**, and Current Sensor **26**
[0042] The relay **12** includes a relay body **52** with a hollow rectangular parallelepiped-like shape. A pair of connection portions **14** and **14** (a first connection portion **14***a* and a second connection portion **14***b*) are provided on the front surface of the relay body **52** separated from one another in the left-and-right direction. An insulating plate **54** projecting frontward is provided between the first connection portion **14***a* and the second connection portion **14***b*.
[0043] Also a plurality of leg portions **56** projecting outward in the left-and-right direction are provided on the relay body **52**. Bolt insertion holes are formed in the leg portions **56** extending through in the up-and-down direction.
[0044] The fuse **24** includes a fuse body **60** with a substantially rectangular parallelepiped-like shape. Connection portions **62** and **62** that are made of metal and project to both sides in the left-and-right direction are provided on the fuse body **60**. Bolt insertion holes are formed in the connection portions **62** and **62** extending through in the up-and-down direction.
[0045] The current sensor **26** includes a sensor body **66** with a substantially rectangular parallelepiped-like shape. Connection portions **68** and **68** that are made of metal and project to both sides in the left-and-right direction are provided on the sensor body **66**. Bolt insertion holes are formed in the connection portions **68** and **68** extending through in the up-and-down direction.
Energization Bus Bar **16**
[0046] The energization bus bar **16** is formed by bending a metal plate material in a predetermined shape via a pressing process or the like. The material of the energization bus bar **16** is not limited, however copper, copper alloy, aluminum, aluminum alloy, or the like are preferably used. Note that the coefficient of linear thermal expansion of copper is approximately 16 to 17 (×10.sup.−6/K). Also, the coefficient of linear thermal expansion of aluminum is approximately 23 to 24 (×10.sup.−6/K). In the first embodiment, as also illustrated in FIG. **4**, the pair of energization bus bars **16** and **16** (first energization bus bar **16***a* and second energization bus bar **16***b*) are provided separated from one another in the left-and-right direction.

[0047] The first energization bus bar **16***a* extends overall in the left-and-right direction. The first energization bus bar **16***a* includes, at the right end portion, a bolt fastening portion **72** with a rectangular shape extending in the up-and-down direction (YZ plane). A heat transfer portion **74** is a rectangular shape extending in the horizontal direction (XY plane) extends from the back of the lower end of the bolt fastening portion **72**. Also, the first energization bus bar **16***a* includes, at the left end portion, an external connection portion **76** with a rectangular shape extending in the horizontal direction (XY plane). Also, the heat transfer portion **74** and the external connection portion **76** are connected by a portion bent like a crank in the intermediate portion in the left-and-right direction.
[0048] Furthermore, the upper end portion of the bolt fastening portion **72** of the first energization bus bar **16***a* is folded forward and overlaps the lower end portion of the bolt fastening portion **72**. Note that the upper end portion of the bolt fastening portion **72** prior to being folded back is indicated by a two-dot dash line in FIG. **4**. The folded back and overlapped portion corresponds to the heat capacity increasing component **20**. In other words, in the first embodiment, the heat capacity increasing component **20** is made of metal and uses the same material as the energization bus bar **16** (first and second energization bus bars **16***a* and **16***b*). The first energization bus bar **16***a* is formed with double thickness at the position where the heat capacity increasing component **20** is formed. Also, the back surface of the bolt fastening portion **72** corresponds to a contact surface **78** that comes into contact with the first connection portion **14***a* of the relay **12**. Thus, the upper end portion (heat capacity increasing component **20**) of the bolt fastening portion **72** corresponding to the end portion of the first energization bus bar **16***a* is overlapped with a front surface **79** corresponding to the surface of the bolt fastening portion **72** on the opposite side to the contact surface **78**.
[0049] Also, the first energization bus bar **16***a* is provided with a holding portion **80** that holds the heat capacity increasing component **20** and the front surface **79** of the bolt fastening portion **72** in an overlapped state. The shape of the holding portion **80** is not limited, however in the first embodiment, the holding portion **80** is a metal member formed integrally with the first energization bus bar **16***a*. Specifically, a pair of band-like holding portions **80** and **80** are provided on the left and right sides of the bolt fastening portion **72**. Also, by bending and crimping the holding portions **80** and **80** with the heat capacity increasing component **20** (upper end portion of the bolt fastening portion **72**) overlapped with the front surface **79** of the bolt fastening portion **72**, the overlapped state of the heat capacity increasing component **20** (upper end portion of the bolt fastening portion **72**) can be held.
[0050] Also, a bolt insertion hole **82** is formed in the bolt fastening portion **72** extending through in the front-and-back direction. In the first embodiment, the bolt insertion hole **82** is formed in the portion where the heat capacity increasing component **20** is provided (the portion where the upper end portion of the bolt fastening portion **72** is folded back and overlapped). Thus, the bolt insertion hole **82** is formed in the portion where the first energization bus bar **16***a* has double thickness extending through in the front-and-back direction. Note that the bolt insertion hole **82** may be formed by forming a through hole in both the upper end portion and the lower end portion of the bolt fastening portion **72** prior to folding back the upper end portion of the bolt fastening portion **72**, folding back the upper end portion of the bolt fastening portion **72**, and connecting the through holes. Alternatively, the bolt insertion hole **82** may be formed at the portion of double thickness after the upper end portion of the bolt fastening portion **72** is folded back and overlapped. By inserting the bolt **18** into the bolt insertion hole **82** and fastening it to the first connection portion **14***a*, the first energization bus bar **16***a* provided with the heat capacity increasing component **20** is affixed. In other words, by fastening the bolt **18**, not only is the first energization bus bar with single thickness fixed to the first connection portion **14***a*, but the heat capacity increasing component **20** also with single thickness is also fixed.
[0051] Also, in the first embodiment, the bolt insertion hole **82** has an elliptical shape elongated in

the up-and-down direction. Thus, when the relay **12** and the first energization bus bar **16***a* are fastened together as described below, the position in the up-and-down direction of the first energization bus bar **16***a* relative to the relay **12** can be adjusted. As a result, as described below, the heat transfer portion **74** can be more reliably thermally brought into contact with the case **22** (or a heat conduction sheet **114** described below). Also, a bolt insertion hole **84** is formed in the external connection portion **76** extending through in the thickness direction (up-and-down direction).

[0052] The second energization bus bar **16***b* has a substantially symmetrical shape in the left-and-right direction to the first energization bus bar **16***a*. That is, the bolt fastening portion **72** is provided at the front of the left end portion of the second energization bus bar **16***b*. The heat transfer portion **74** extends backward from the lower end portion of the bolt fastening portion **72**. Also, a fuse connection portion **86** with a rectangular shape extending in the horizontal direction (XY plane) is provided on the right end portion of the second energization bus bar **16***b*. The heat transfer portion **74** and the fuse connection portion **86** are connected by a portion bent like a crank in the intermediate portion in the left-and-right direction. Also, a bolt insertion hole **88** is formed in the fuse connection portion **86** extending through in the thickness direction (up-and-down direction).

[0053] Furthermore, the upper end portion of the bolt fastening portion **72** of the second energization bus bar **16***b* is folded forward, forming the heat capacity increasing component **20**. Also, the state of the heat capacity increasing component **20** (upper end portion of the bolt fastening portion **72**) being folded and overlapped with the front surface **79** is held by the holding portions **80** and **80**. Also, in a state in which the heat capacity increasing component **20** is provided, the bolt insertion hole **82** is formed in the bolt fastening portion **72** extending through in the thickness direction (front-and-back direction). By inserting the bolt **18** into the bolt insertion hole **82** and fastening it to the second connection portion **14***b*, the second energization bus bar **16***b* provided with the heat capacity increasing component **20** is affixed. In other words, by fastening the bolt **18**, the heat capacity increasing component **20** together with the second energization bus bar **16***b* is fixed to the second connection portion **14***b*.

Third Energization Bus Bar **90** and Fourth Energization Bus Bar **92**

[0054] As illustrated in FIGS. **2** and **3**, a third energization bus bar **90** is connected to the fuse **24** and the current sensor **26**. Also, at the current sensor **26**, a fourth energization bus bar **92** is connected on the opposite side to the side where the third energization bus bar **90** is connected. The third and fourth energization bus bars **90** and **92** are also formed by bending a metal plate material in a predetermined shape via a pressing process or the like as with the first and second energization bus bars **16***a* and **16***b*.

[0055] The third energization bus bar **90** includes, on the left and right end portions, a fuse connection portion **94** and a sensor connection portion **96** with a rectangular shape extending in the horizontal direction. That is, on the left side of the third energization bus bar **90**, the fuse connection portion **94** is provided, and on the right side, the sensor connection portion **96** is provided. A bolt insertion hole is formed in both the fuse connection portion **94** and the sensor connection portion **96** extending through in the thickness direction (up-and-down direction).

[0056] In the first embodiment, the third energization bus bar **90** includes a

[0057] substantially trough-shaped portion extending in the front-and-back direction and opening upward. The fuse connection portion **94** and the sensor connection portion **96** extends outward in the left and right direction from the left and right end portions of the upper opening portion of the substantially trough-shaped portion. Also, the bottom wall of the substantially trough-shaped portion corresponds to a heat transfer portion **102** thermally in contact with the case **22** (base member **28**) when the circuit assembly **10** is assembled.

[0058] The fourth energization bus bar **92** has a structure similar to that of the third energization bus bar **90**. In other words, the fourth energization bus bar **92** includes a substantially trough-shaped portion extending in the front-and-back direction and opening upward. A sensor connection portion **104** extends to the left from the left end portion of the upper opening portion of the substantially trough-shaped portion, and an external connection portion **106** extends to the right from the right end portion. A bolt insertion hole is formed in the sensor connection portion **104** extending through in the thickness direction (up-and-down direction). Also, a bolt insertion hole **110** is formed in the external connection portion **106** extending through in the thickness direction (up-and-down direction). Also, the bottom wall of the substantially trough-shaped portion corresponds to a heat transfer portion **112** thermally in contact with the case **22** (base member **28**) when the circuit assembly **10** is assembled.

Bolt **18**

[0059] The relay **12** and the first and second energization bus bars **16***a* and **16***b* are fixed together via the bolts **18** and **18**. Specifically, the first and second connection portions **14***a* and **14***b* and the bolt insertion holes **82** and **82** of the bolt fastening portions **72** and **72** are aligned in position, and the bolts **18** and **18** are inserted and fastened. A known material, such as iron or stainless steel, may be used as the material of the bolt **18**. In the first embodiment, the bolt **18** is made of iron. Note that the coefficient of linear thermal expansion of iron is approximately 11 to 12 (×10.sup.−6/K).

Heat Conduction Sheets **114** and **116**

[0060] When the circuit assembly **10** is assembled, the heat transfer portions **74**, **74**, **102**, and **112** of the first to fourth energization bus bars **16***a*, **16***b*, **90**, and **92** are thermally in contact with the case **22** (base member **28**). In the first embodiment, the heat conduction sheets **114**, i.e., heat conducting members, are housed in the first housing recess portions **36** of the base member **28**. Also, the heat transfer portions **74**, **74**, **102**, and **112** are thermally in contact with the base member **28** via the heat conduction sheets **114**.

[0061] Also, in the first embodiment, heat conduction sheets **116** are housed in the second housing recess portions **40** of the base member **28**. When the circuit assembly **10** is installed in a vehicle, the base member **28** is thermally in contact with a heat dissipating body **118**, such as a vehicle body panel or housing, via the heat conduction sheets **116**.

[0062] The heat conduction sheets **114** and **116** have a sheet-like shape flat in the up-and-down direction are made of a synthetic resin with a larger thermal conductivity than air. Specifically, a silicone-based resin, a non-silicone-based acrylic resin, a ceramic-based resin, or the like can be used. A more specific example is heat-conductive silicone rubber. The heat conduction sheets **114** and **116** have flexibility and elasticity and can elastically deform, changing in the thickness dimension, in response to a force applied in the up-and-down direction. Note that in the first embodiment, the heat conduction sheets **114** and **116** are used as heat conducting members provided on the upper and lower surfaces of the base member **28**. However, the heat conducting members are in no way limited to this configuration. The heat conducting members may have a discretionary shape, and a heat dissipating gap filler or heat conducting grease made of a silicone-based resin may be used, for example.

[0063] In particular, in the first embodiment, because the heat conduction sheets **114** are housed in the first housing recess portions **36** with the level difference **38**, the heat conduction sheets **114** are positioned relative to the base member **28**. Also, because the heat conduction sheets **116** are housed in the second housing recess portions **40** with the level difference **42**, the heat conduction sheets **116** are positioned relative to the base member **28**. Furthermore, the heat conduction sheets **114** are preferably sandwiched between the heat transfer portions **74**, **74**, **102**, and **112** and the base member **28** in a compressed state in the up-and-down direction.

[0064] By compressing the heat conduction sheets **114**, the heat transfer portions **74**, **74**, **102**, and **112** and the base member **28** can be brought into contact with a high degree of adhesion. Accordingly, the heat conduction sheets **114** can efficiently transfer heat from the heat transfer portions **74**, **74**, **102**, and **112** to the base member **28**. In a similar manner, the heat conduction sheets **116** are preferably sandwiched between the base member **28** and the heat dissipating body

118 in a compressed state in the up-and-down direction. By compressing the heat conduction sheets **116**, the base member **28** and the heat dissipating body **118** can be brought into contact with a high degree of adhesion. Accordingly, the heat conduction sheets **116** can efficiently transfer heat from the base member **28** to the heat dissipating body **118**.

Assembly Process of Circuit Assembly **10**

[0065] Next, a detailed example of the assembly process of the circuit assembly **10** will be described. Note that the assembly process of the circuit assembly **10** is not limited to that described below.

[0066] First, the lid member **30**, the relay **12**, the fuse **24**, the current sensor **26**, the first to fourth energization bus bars **16***a*, **16***b*, **90**, and **92**, and the bolts **18** are prepared. Then, the relay **12** is placed against the upper bottom wall **44** of the lid member **30** inverted upside down and the bolts are inserted in the leg portions **56** and fastened in the not-illustrated bolt fixing portions provided in the lid member **30**. This fixes the lid member **30** and the relay **12** together. Then, the first and second energization bus bars **16***a* and **16***b* are placed above the relay **12**, and the first and second connection portions **14***a* and **14***b* of the relay **12** and the bolt insertion holes **82** and **82** of the first and second energization bus bars **16***a* and **16***b* are aligned in position. Then, the bolts **18** and **18** are inserted in the first and second connection portions **14***a* and **14***b* and the bolt insertion holes **82** and **82** to fasten them together. In this manner, the relay **12** and the first and second energization bus bars **16***a* and **16***b* are fixed together.

[0067] Next, the third energization bus bar **90** and the fourth energization bus bar **92** are placed against the upper bottom wall **44** of the lid member **30**, and the fuse **24** and the current sensor **26** are also placed from above. In this manner, the fuse connection portion **86** of the second energization bus bar **16***b* and the connection portion **62** on the left side of the fuse **24** are aligned and overlapped. Also, the connection portion **62** on the right side of the fuse **24** and the fuse connection portion **94** of the third energization bus bar **90** are aligned and overlapped. The sensor connection portion **96** of the third energization bus bar **90** and the connection portion **68** on the left side of the current sensor **26** are aligned and overlapped. Also, the connection portion **68** on the right side of the current sensor **26** and the sensor connection portion **104** of the fourth energization bus bar **92** are aligned and overlapped. Then, the bolts are inserted in the overlapped connection portions **62** and **68**, the fuse connection portions **86** and **94**, and the sensor connection portions **96** and **104** and fastened to not-illustrated bolt fixing portions provided on the lid member **30**. In this manner, in addition to the relay **12** and the first and second energization bus bars **16***a* and **16***b*, the fuse **24**, the current sensor **26**, the third energization bus bar **90**, and the fourth energization bus bar **92** are fixed to the lid member **30**.

[0068] The base member **28** and the heat conduction sheets **114** and **116** are also prepared. Then, the heat conduction sheets **114** are housed in the first housing recess portions **36** of the base member **28** and fixed via adhesive or the like. The heat conduction sheets **116** are also housed in the second housing recess portions **40** and fixed via adhesive or the like. Then, the upper opening portion of the lid member **30** to which the relay **12**, the fuse **24**, the current sensor **26**, and the first to fourth energization bus bars **16***a*, **16***b*, **90**, and **92** are fixed is covered with the base member **28** to which the heat conduction sheets **114** and **116** are fixed, and the lid member **30** and the base member **28** are fixed together to form the case **22**. Then, by inverting it, the circuit assembly **10** is completed.

[0069] Note that the order of fixing the relay **12**, the fuse **24**, the current sensor **26**, and the first to fourth energization bus bars **16***a*, **16***b*, **90**, and **92** to the lid member **30** is not limited to that described in the process described above. Also, the heat conduction sheets **114** provided between the first to fourth energization bus bars **16***a*, **16***b*, **90**, and **92** (heat transfer portions **74**, **74**, **102**, and **112**) and the base member **28** may be fixed to the lower surface of the heat transfer portions **74**, **74**, **102**, and **112** and not fixed to the base member **28**. In a similar manner, the heat conduction sheets **116** provided on the lower surface of the base member **28** may be fixed to the heat dissipating body

118 and not fixed to the base member **28**.
[0070] In the circuit assembly **10** assembled in this manner, the external connection portions **76** and **106** of the first energization bus bar **16***a* and the fourth energization bus bar **92** are exposed to the outside via the opening portions **48***a* and **48***b* of the lid member **30**. Then, in a state where terminal portions provided on terminals of not-illustrated external electrical wires and the bolt insertion holes **84** and **110** of the external connection portions **76** and **106** are aligned in position and not-illustrated bolts are inserted and fastened, the external electrical wire and the first energization bus bar **16***a* and the fourth energization bus bar **92** are electrically connected. Also, by overlapping the circuit assembly **10** and the heat dissipating body **118** and inserting and fastening not-illustrated bolts in the bolt insertion holes **50** provided in the outer peripheral portion of the case **22** (lid member **30**), the circuit assembly **10** is fixed to the heat dissipating body **118**. In this manner, in the first embodiment, the heat conduction sheets **116** are compressed between the circuit assembly **10** and the heat dissipating body **118** in the up-and-down direction.
[0071] The circuit assembly **10** of the first embodiment is provided with the heat capacity increasing components **20** and **20** that are thermally in contact with the fastening sites A of the first and second connection portions **14***a* and **14***b* of the relay **12** at the first and second energization bus bars **16***a* and **16***b*. Specifically, by folding back and overlapping the upper end portion of the bolt fastening portions **72** and **72** of the first and second energization bus bars **16***a* and **16***b*, the heat capacity increasing component **20** is formed. Accordingly, at the fastening sites A of the first and second energization bus bars **16***a* and **16***b* and the first and second connection portions **14***a* and **14***b* of the relay **12**, the first and second energization bus bars **16***a* and **16***b* have double thickness. Thus, compared to a case in which the first and second energization bus bars have simply single thickness, the heat capacity of the first and second energization bus bars **16***a* and **16***b* can be increased. In this manner, an increase in the temperature of the first and second energization bus bars **16***a* and **16***b* and thus the first and second connection portions **14***a* and **14***b* that connect to the first and second energization bus bars **16***a* and **16***b* can be suppressed, and the problem of heat generation when a large current temporarily flows can be solved.
[0072] Also, in the first embodiment, the heat transfer portions **74**, **74**, **102**, and **112** of the first to fourth energization bus bars **16***a*, **16***b*, **90**, and **92** are each thermally in contact with the case **22** (base member **28**). Thus, the heat generated at the relay **12**, the fuse **24**, and the current sensor **26** when energized can be dissipated via the case **22**. Accordingly, the problem of heat generated by the relay **12**, the fuse **24**, and the current sensor **26** can be solved. In particular, in the first embodiment, the heat conduction sheets **114** are provided between the heat transfer portions **74**, **74**, **102**, and **112** and the base member **28**. Thus, a stable transfer of heat from the heat transfer portions **74**, **74**, **102**, and **112** to the base member **28** can be achieved.
[0073] Furthermore, the heat conduction sheets **116** are provided on the lower surface of the base member **28**. Thus, the base member **28** and the heat dissipating body **118** are thermally in contact with each other via the heat conduction sheets **116**. Accordingly, the heat generated at the relay **12**, the fuse **24**, and the current sensor **26** is dissipated also from the heat dissipating body **118**, allowing the heat dissipation effect to be improved.
[0074] Also, in the first embodiment, the heat capacity increasing components **20** and **20** are formed by the upper end portions of the bolt fastening portions **72** and **72** of the first and second energization bus bars **16***a* and **16***b*, and the bolt insertion holes **82** and **82** are formed on the portions where the heat capacity increasing components **20** and **20** are provided. Accordingly, by fastening the bolts **18** and **18**, the heat capacity increasing components **20** and **20** are fixed together with the first and second energization bus bars **16***a* and **16***b*. In other words, in the first embodiment, the first and second energization bus bars **16***a* and **16***b* and the heat capacity increasing components **20** and **20** are integrally formed. This allows an increase in the number of parts to be avoided. Also, compared to a case in which the heat capacity increasing component is a separate member from the first and second energization bus bars **16***a* and **16***b*, the ease of assembly

can be improved.
[0075] In particular, in the first embodiment, the upper end portions of the bolt fastening portions **72** and **72** are folded outward (forward) and overlapped. Accordingly, compared to a case in which the upper end portions of the bolt fastening portions are folded inward, the electrical path from the first and second connection portions **14***a* and **14***b* to the external connection portion **76** and the fuse connection portion **86** can be shortened. This allows an increase to the electrical conduction resistance when energized to be avoided. Also, the thermal path from the first and second connection portions **14***a* and **14***b* to the heat transfer portions **74** and **74** can be shortened. Accordingly, the heat generated at the first and second connection portions **14***a* and **14***b* is quickly dissipated via the heat transfer portions **74** and **74**.
[0076] Also, the first and second energization bus bars **16***a* and **16***b* are provided with the holding portions **80** and **80** that hold the heat capacity increasing components **20** and **20** (upper end portions of the bolt fastening portions **72** and **72**) in an overlapped state. Thus, no gap is formed between heat capacity increasing components **20** and **20** and the bolt fastening portions **72** and **72**, i.e., between the upper end portions and the lower end portions of the bolt fastening portions **72** and **72** overlapping one another, and the heat capacity at the position where the heat capacity increasing components **20** and **20** are provided can be stably increased.
[0077] Note that the coefficient of linear thermal expansion of the heat capacity increasing component **20** (first and second energization bus bars **16***a* and **16***b*) is preferably set within a range of from ⅓ to 3 times the coefficient of linear thermal expansion of the bolts **18** and **18**. The coefficient of linear thermal expansion of the heat capacity increasing component **20** (first and second energization bus bars **16***a* and **16***b*) is more preferably set within a range of from ½ to 2 times the coefficient of linear thermal expansion of the bolts **18** and **18**. The coefficient of linear thermal expansion of the heat capacity increasing component **20** (first and second energization bus bars **16***a* and **16***b*) is even more preferably set within a range of from ⅔ to 3/2 times the coefficient of linear thermal expansion of the bolts **18** and **18**. The coefficient of linear thermal expansion of the heat capacity increasing component **20** (first and second energization bus bars **16***a* and **16***b*) is most preferably set to be equal to the coefficient of linear thermal expansion of the bolts **18** and **18**. By setting the coefficient of linear thermal expansion of the heat capacity increasing component **20** (first and second energization bus bars **16***a* and **16***b*) to from ⅓ to 3 times, i.e., relatively close to, the coefficient of linear thermal expansion of the bolts **18** and **18**, looseness of the bolts **18** and **18** when heat is generated at the relay **12** can be suppressed. Note that, for example, in a case in which the first and second energization bus bars **16***a* and **16***b* are made of copper and the bolts **18** and **18** are made of iron, the coefficient of linear thermal expansion of the heat capacity increasing components **20** and **20** is approximately **1.4** times the coefficient of linear thermal expansion of the bolts **18** and **18**.
[0078] In particular, because the coefficient of linear thermal expansion of the heat capacity increasing component **20** (first and second energization bus bars **16***a* and **16***b*) and the bolts **18** and **18** are equal, i.e., the heat capacity increasing component **20** (first and second energization bus bars **16***a* and **16***b*) and the bolts **18** and **18** are made of the same material, looseness of the bolts **18** and **18** when heat is generated at the relay **12** can be further suppressed.
Second Embodiment
[0079] The second embodiment of the present disclosure will be described below with reference to FIGS. **6** and **7**. A circuit assembly **120** of the second embodiment has basically a similar configuration to the circuit assembly **10** of the first embodiment but is different in that heat capacity increasing components **122** and **122** are separate members from first and second energization bus bars **124***a* and **124***b*, which are energization members. Hereinafter, the members and portions that are essentially the same as those in the embodiment described above are given the same reference sign as in the embodiment described above and a detailed description thereof will be omitted. Note that FIGS. **6** and **7** are diagrams illustrating the circuit assembly **120** in a state

with the lid member **30** that forms the case **22** removed.

[0080] The heat capacity increasing component **122** according to the second embodiment has a rectangular block shape. A through hole **126** is formed in a substantially central portion of the heat capacity increasing component **122** extending through in the front-and-back direction. The material of the heat capacity increasing component **122** is not limited and can be any material that is capable of increasing the heat capacity of the first and second energization bus bars **124***a* and **124***b* and thus the first and second connection portions **14***a* and **14***b* when assembled. A metal with a high thermal conductivity is preferably used. Iron, copper, aluminum, an alloy thereof, or the like is more preferably used as the material of the heat capacity increasing component **122**. In the second embodiment, the heat capacity increasing component **122** is made of metal. Note that the heat capacity increasing component **122** is preferably made of a metal with a lighter specific gravity than copper and iron. This is because a metal with a light specific gravity can reduce the effects of vibration in the bolt **18**.

[0081] The heat capacity increasing components **122** of the second embodiment are fixed to the relay **12** together with the first and second energization bus bars **124***a* and **124***b* via the bolts **18**. In other words, the first and second connection portions **14***a* and **14***b* of the relay **12**, the bolt insertion holes **82** of the first and second energization bus bars **124***a* and **124***b*, and the through holes **126** of the heat capacity increasing components **122** are aligned in position with one another. Then, by inserting and fastening the bolts **18**, the heat capacity increasing components **122** and **122** are fixed to the relay **12** together with the first and second energization bus bars **124***a* and **124***b*. Accordingly, the heat capacity increasing components **122** and **122** are thermally in contact with the fastening sites A of the first and second energization bus bars **124***a* and **124***b* and the first and second connection portions **14***a* and **14***b*. In particular, in the second embodiment also, the heat capacity increasing component **122** is overlapped with the front surface **79**, which is the surface on the opposite side to the contact surface **78** with the first and second connection portions **14***a* and **14***b* at the bolt fastening portion **72** of the first and second energization bus bars **124***a* and **124***b*. Note that in the first embodiment, the upper end portion of the bolt fastening portion **72** is folded back, and the fastening site of the bolt **18** at the first and second energization bus bars **16***a* and **16***b* have double thickness. However, the fastening site of the bolt **18** on the first and second energization bus bars **124***a* and **124***b* of the second embodiment have single thickness.

[0082] In the circuit assembly **120** of the second embodiment also, because the heat capacity increasing component **122** is provided, the heat capacity at the fastening site A of the first and second connection portions **14***a* and **14***b* of the relay **12** and the first and second energization bus bars **124***a* and **124***b* is increased. Thus, heat generation at the relay **12** can be suppressed. Thus, a similar effect to that of the first embodiment is achieved.

[0083] In particular, in the circuit assembly **120** of the second embodiment, the heat capacity increasing component **122** is a separate member from the first and second energization bus bars **124***a* and **124***b*. Thus, for the material of the heat capacity increasing component **122**, a material that tends to increase the heat capacity more than the first and second energization bus bars **124***a* and **124***b* can be used. Alternatively, for the material of the heat capacity increasing component **122**, the same material (for example, iron) as the bolt **18** may be used, allowing looseness of the bolt **18** when heat is generated at the relay **12** to be reduced. The shape of the heat capacity increasing component **122** is not limited to a rectangular block shape, and a simple flat plate shape like a bus bar, a shape that tends to increase the heat capacity, or a shape capable of suppressing looseness of the bolt **18** when heat is generated may be used.

[0084] Also, in the second embodiment also, the heat capacity increasing component **122** is provided overlapping the front surface **79** of the bolt fastening portion **72** of the first and second energization bus bars **124***a* and **124***b*. Thus, the electrical path from the first and second connection portions **14***a* and **14***b* to the external connection portion **76** and the fuse connection portion **86** and the thermal path to the heat transfer portions **74** and **74** can be shortened, an increase in the electrical conduction resistance can be prevented, and heat can be quickly transferred.

Third Embodiment

[0085] The third embodiment of the present disclosure will be described below with reference to FIGS. **8** and **9**. A circuit assembly **130** of the third embodiment has basically a similar configuration to the circuit assembly **120** of the second embodiment but is different in that, instead of the heat capacity increasing component **122**, caps **132** and **132** made of metal are installed on the bolts **18** as heat capacity increasing components. Note that in the third embodiment, the first and second energization bus bars **124***a* and **124***b* with a similar structure to that of the second embodiment are used. Also, FIGS. **8** and **9** are diagrams illustrating the circuit assembly **130** in a state with the lid member **30** that forms the case **22** removed.

[0086] That is, in the third embodiment, the heat capacity increasing component is formed by the cap **132** installed on the head portion of the bolt **18**. Thus, a housing recess portion **136** that houses the head portion of the bolt **18** is formed in the cap **132**. Accordingly, the cap **132** is thermally in contact with the fastening site A of the first and second energization bus bars **124***a* and **124***b* and the first and second connection portions **14***a* and **14***b* via the bolt **18**. In the third embodiment, the cap **132** is made of a metal with a high thermal conductivity. The cap **132** is preferably made of iron, copper, aluminum, an alloy thereof, or the like, for example. That is, the heat capacity increasing component is not limited to a configuration in which it is fixed to a heat generating member (relay **12**) together with an energization member (first and second energization bus bar) via a fastening member (bolt **18**).

[0087] The caps **132** are installed on the head portions of the bolts **18** after the first and second energization bus bars **124***a* and **124***b* are fixed to the first and second connection portions **14***a* and **14***b* of the relay **12** via the bolts **18**. Alternatively, the first and second energization bus bars **124***a* and **124***b* may be fixed to the first and second connection portions **14***a* and **14***b* of the relay **12** via the bolts **18** after the caps **132** are installed on the head portions of the bolts **18**. A portion of the cap **132** on the outer peripheral side of the housing recess portion **136** may be in contact with the front surface **79** of the bolt fastening portion **72** of the first and second energization bus bars **124***a* and **124***b* or may not be in contact.

[0088] Note that preferably, a heat conducting grease **138**, which is a heat conducting member, is provided between the inner surface of the housing recess portion **136** of the cap **132** and the head portion of the bolt **18**. In this manner, even in a case in which a manufacturing error or the like causes a gap to be formed between the cap **132** and the bolt **18**, heat can be stably transferred from the bolt **18** to the cap **132**.

[0089] In the circuit assembly **130** of the third embodiment, the caps **132** that increase the heat capacity of the first and second connection portions **14***a* and **14***b* are provided on the bolts **18** that fix together the relay **12** and the first and second energization bus bars **124***a* and **124***b*. As a result, an increase in the temperature of the bolts **18** and thus the first and second connection portions **14***a* and **14***b* when heat is generated at the relay **12** can be suppressed by the caps **132**.

[0090] In particular, in the third embodiment, because the caps **132** are made of metal, the heat capacity can be easily increased. Also, because the caps **132** are separate members from the first and second energization bus bars **124***a* and **124***b* and the bolts **18**, for the material of the caps **132**, a material that tends to increase the heat capacity more than the first and second energization bus bars **124***a* and **124***b* and the bolts **18** can be used. Note that by making the cap **132** and the bolt **18** have a similar coefficient of linear thermal expansion or by the cap **132** and the bolt **18** being the same material, a gap can be made unlikely to form between the cap **132** and the bolt **18** when heat is generated at the relay **12**. Note that the cap **132** is preferably made of a metal with a lighter specific gravity than copper and iron. This is because a metal with a light specific gravity can reduce the effects of vibration in the bolt **18**.

Fourth Embodiment

[0091] The fourth embodiment of the present disclosure will be described below with reference to

FIG. **10**. A circuit assembly **140** of the fourth embodiment has basically a similar configuration to the circuit assembly **10** of the first embodiment but is different in that a cap **142** made of synthetic resin is installed on the bolt **18** as a heat capacity increasing component. In other words, the cap **142** is thermally in contact with the fastening site A of the first and second energization bus bars **16***a* and **16***b* and the first and second connection portions **14***a* and **14***b* via the bolt **18**. Note that FIG. **10** is a diagram illustrating the circuit assembly **140** in a state with the lid member **30** that forms the case **22** removed.
[0092] In the circuit assembly **140** of the fourth embodiment, the cap **142** that increases the heat capacity of the first and second connection portions **14***a* and **14***b* is provided on the bolt **18** that fixes together the relay **12** and the first and second energization bus bars **16***a* and **16***b*. Thus, an effect of suppressing an increase in the temperature from the cap **142** is added to that of the heat capacity increasing component **20** of the first embodiment. In particular, in the fourth embodiment, because synthetic resin which is relatively softer than metal is used for the cap **142**, the head portion of the bolt **18** and the cap **142** can be brought into close contact substantially without gaps, allowing for the contact area between the cap **142** and the head portion of the bolt **18** to be substantially ensured. In this manner, heat can be stably transferred from the bolt **18** to the cap **142**. Furthermore, because the cap **142** made of synthetic resin is used, electrical insulating properties at the head portion of the bolt **18** is ensured.
Other Embodiments
[0093] The technology described in the present specification is not limited to the embodiments described above with reference to the drawings, and, for example, the following embodiments are also included in the technical scope of the technology described in the present specification. [0094] (1) In the embodiments, the heat capacity increasing components **20** and **122**
[0095] and the caps **132** and **142** are provided at the fastening site A of the first and second connection portions **14***a* and **14***b* of the relay **12**, which is the heat generating component, and the first and second energization bus bars **16***a*, **124***a*, **16***b*, and **124***b*. However, no such limitation is intended. The heat capacity increasing component may be provided at a fastening site of the connection portion of the fuse or current sensor that generates heat when energized and the energization member (for example, the second to fourth energization bus bars in the embodiments). In other words, the heat generating component according to the present disclosure may be a fuse or a current sensor instead of or in addition to the relay. Note that the number of heat generating components provided is not necessarily a plurality and it is sufficient that one or more are provided.
[0096] (2) In the third embodiment, the cap **132** is used as the heat capacity increasing component instead of the heat capacity increasing component **20** of the first embodiment. However, the cap **132** may be used in addition to the heat capacity increasing components **20** and **122** of the first and second embodiments. [0097] (3) The heat capacity increasing components **20** and **122** and the caps **132** and **142** of the embodiments, in a configuration other than that of the fourth embodiment, may be used in a combination of two or more. In other words, the heat capacity increasing component of the first and second embodiments combined may be used at the fastening site of the first and second connection portions of the relay and the first and second energization bus bars, for example. Alternatively, the heat capacity increasing component of the first and second embodiment may be used at the fastening site of the first and second connection portions of the relay and the first and second energization bus bars together with the heat capacity increasing component of the third and fourth embodiment at the fastening site of the connection portion of a fuse or current sensor and the energization member. [0098] (4) In the embodiments, the heat capacity increasing components **20** and **122** and the caps **132** and **142** as heat capacity increasing components are provided at the fastening sites A of the first and second connection portions **14***a* and **14***b* of the relay **12** and the first and second energization bus bars **16***a*, **124***a*, **16***b*, and **124***b*.
[0099] However, no such limitation is intended. It is sufficient that the heat capacity increasing component is provided at at least one fastening site of the connection portion and the energization

member. Note that this also applies to a case in which the heat capacity increasing component is provided at the fastening site of the connection portion of a fuse or a current sensor and the energization member. [0100] (5) A heat dissipation mechanism (for example, the heat transfer portions **74**, **102**, and **112**, heat conduction sheets **114** and **116**, and the like) for dissipating heat from the component (for example, the relay **12**, the fuse **24**, and the current sensor **26** in the embodiments) that generates heat when energized is not necessary. Even in a case in which the heat dissipation mechanism is provided, the structure is not limited to that described in the embodiments, and a known heat dissipation mechanism may be used. For example, a through hole may be provided in the case (the bottom wall of the base member, for example), and the heat transfer portion may be thermally in contact with the heat dissipating body directly or via the heat conducting member (the heat conduction sheet, for example). [0101] (6) In the embodiments, the relay **12**, the fuse **24**, the current sensor **26**, and the first to fourth energization bus bars **16***a*, **124***a*, **16***b*, **124***b*, **90**, and **92** are all fixed to the lid member **30**, but one or more may be fixed to the base member. [0102] (7) In the embodiments, the bolt **18** is used as an example of the fastening member. However, the fastening member is not limited to the bolt, and a known fastening member such as a rivet capable of fastening together the energization member and the connection portion can be used. [0103] (8) The heat capacity increasing component according to the present disclosure is not limited to the shape and material described in the embodiments, and the shape and material are not limited as long as that by providing the heat capacity increasing component, the heat capacity increases more than in the case of a single energization member.

LIST OF REFERENCE NUMERALS

[0104] **10** Circuit assembly (first embodiment) [0105] **12** Relay (heat generating component) [0106] **14** Connection portion [0107] **14***a* First connection portion [0108] **14***b* Second connection portion [0109] **16** Energization bus bar (energization member) [0110] **16***a* First energization bus bar [0111] **16***b* Second energization bus bar [0112] **18** Bolt (fastening member) [0113] **20** Heat capacity increasing component [0114] **22** Case [0115] **24** Fuse [0116] **26** Current sensor [0117] **28** Base member [0118] **30** Lid member [0119] **32** Bottom wall [0120] **34** Peripheral wall [0121] **36** First housing recess portion [0122] **38** Level difference [0123] **40** Second housing recess portion [0124] **42** Level difference [0125] **44** Upper bottom wall [0126] **46** Peripheral wall [0127] **48***a*, **48***b* Opening portion [0128] **50** Bolt insertion hole [0129] **52** Relay body [0130] **54** Insulating plate [0131] **56** Leg portion [0132] **60** Fuse body [0133] **62** Connection portion [0134] **66** Sensor body [0135] **68** Connection portion [0136] **72** Bolt fastening portion [0137] **74** Heat transfer portion [0138] **76** External connection portion [0139] **78** Contact surface [0140] **79** Front surface (surface on opposite side to contact surface) [0141] **80** Holding portion [0142] **82**, **84** Bolt insertion hole [0143] **86** Fuse connection portion [0144] **88** Bolt insertion hole [0145] **90** Third energization bus bar [0146] **92** Fourth energization bus bar [0147] **94** Fuse connection portion [0148] **96** Sensor connection portion [0149] **102** Heat transfer portion [0150] **104** Sensor connection portion [0151] **106** External connection portion [0152] **110** Bolt insertion hole [0153] **112** Heat transfer portion [0154] **114** Heat conduction sheet (heat conducting member) [0155] **116** Heat conduction sheet [0156] **118** Heat dissipating body [0157] **120** Circuit assembly (second embodiment) [0158] **122** Heat capacity increasing component [0159] **124***a* First energization bus bar (energization member) [0160] **124***b* Second energization bus bar (energization member) [0161] **126** Through hole [0162] **130** Circuit assembly (third embodiment) [0163] **132** Cap (heat capacity increasing component) [0164] **136** Housing recess portion [0165] **138** Heat conducting grease (heat conducting member) [0166] **140** Circuit assembly (fourth embodiment) [0167] **142** Cap (heat capacity increasing component) [0168] A Fastening site

Claims

1. A circuit assembly comprising: a heat generating component that generates heat when energized; an energization member that connects to a connection portion of the heat generating component; a fastening member that fastens the energization member to the connection portion; and a heat capacity increasing component that is thermally in contact with a fastening site of the energization member and the connection portion and increases a heat capacity of the connection portion of the heat generating component.
2. The circuit assembly according to claim 1, further comprising: a heat conducting member and a case, wherein the energization member is thermally in contact with the case via the heat conducting member.
3. The circuit assembly according to claim 1, wherein the heat capacity increasing component is made of metal; and the heat capacity increasing component is fastened to the connection portion together with the energization member via the fastening member.
4. The circuit assembly according to claim 3, wherein the heat capacity increasing component is overlapped with a surface on an opposite side to a contact surface of the energization member with the connection portion.
5. The circuit assembly according to claim 3, wherein the heat capacity increasing component is formed of an end portion of the energization member and is folded back and overlapped with a surface on an opposite side to a contact surface of the energization member with the connection portion.
6. The circuit assembly according to claim 5, wherein the energization member includes a holding portion that holds the end portion of the energization member forming the heat capacity increasing component and the surface on the opposite side of the contact surface of the energization member with the connection portion in an overlapped state.
7. The circuit assembly according to claim 3, wherein a coefficient of linear thermal expansion of the heat capacity increasing component ranges from ⅓ to 3 times a coefficient of linear thermal expansion of the fastening member.
8. The circuit assembly according to claim 3, wherein the heat capacity increasing component and the fastening member are made of a same material.
9. The circuit assembly according to claim 1, wherein the heat capacity increasing component is formed of a cap configured to be installed on the fastening member.
10. The circuit assembly according to claim 9, wherein the cap is made of metal.
11. The circuit assembly according to claim 10, wherein a heat conducting member is provided between the cap and the fastening member.
12. The circuit assembly according to claim 9, wherein the cap is made of synthetic resin.